DO NOW — RECALL

Sequence & branching in C#

Name: _____ Class: _____ Date: _____

1. Why would Unicode be chosen instead of standard ASCII for a multilingual application?

2. Convert 01011000 0011 to denary using an 8-bit two’s-complement mantissa and a 4-bit two’s-complement exponent.

3. Convert $6D_{16}$ to eight-bit binary.

4. Convert 11110110 from 8-bit two’s-complement binary to denary. Show signed place values.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Unicode supports a much larger repertoire of characters from many writing systems. Standard ASCII has 128 codes and is centred on a limited character set. Unicode provides code points for many languages and symbols.

2. Model answer

5.5 Mantissa 01011000 = 0.6875 ; exponent 0011 = 3 ; $0.6875 \times 2^3 = 5.5$.

3. Model answer

01101101_2 6 → 0110 and D → 1101 .

4. Model answer

−10 −128 + 64 + 32 + 16 + 4 + 2 = −10 .

By the end, you can…

• identify sequence and branching as programming constructs;

• write input, output and simple casts using the C# syntax in your .NET Fiddle guide;

• write and test if / else if / else decisions using suitable conditions.

Vocabulary — know these terms

Recall the term — Sequence

Recall the term — Branching / selection

Recall the term — Boolean expression

Recall the term — Condition

Recall the term — Relational operators

Recall the term — Logical operators

Recall the term — Parsing

Recall the term — Boundary test data

NEW LEARNING

Read, annotate and explain

Use the guide syntax

```
Console.Write("Prompt: ");
string text = Console.ReadLine();

int number =
int.Parse(Console.ReadLine());

if (condition)
{
// instructions
}
else if (condition)
{
// instructions
}
else
{
// instructions
}
```

OCR constructs: sequence · branching/selection · iteration. A past OCR question expects the construct name and a working example.

Sequence = instructions in order

Sequence is the normal top-to-bottom execution of statements. Each statement happens after the previous one unless another construct changes the flow.

Input arrives as text

Console.ReadLine() returns a string. If the program needs a whole number, parse the input into an int .

Branching chooses a route

An if condition must evaluate to true or false . Only the matching branch is executed.

Conditions and order matter

Use == to compare equality, not = . Use && , || and ! for AND, OR and NOT.

With else if , conditions are tested from top to bottom. Put the most demanding threshold first.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Build a result checker

Use the standard .NET Fiddle structure

```
using System;

public class Program
{
    public static void Main()
    {
        // code goes here
    }
}
```

Main program instructions belong inside Main() .

Read the score as an integer

```
Console.Write("Enter score: ");
int score =
    int.Parse(Console.ReadLine());
```

Console.ReadLine() gives us text; int.Parse(...) converts it to a whole number.

Test the highest threshold first

```
if (score >= 70)
{
    Console.WriteLine("Distinction");
}
else if (score >= 50)
{
    Console.WriteLine("Pass");
}
else
{
    Console.WriteLine("Not yet");
}
```

A score of 85 satisfies both >= 70 and >= 50 , so the higher threshold must be tested first.

Use values that expose mistakes

```
85 → Distinction
70 → Distinction
50 → Pass
49 → Not yet
```

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY: Even or odd

Ask for a whole number. Output Even when the remainder after division by 2 is zero; otherwise output Odd .

```
Console.Write("Number: ");
int number = int.Parse(Console.ReadLine());

// ?
```

Method / working:

Guided 2 — TRY: Three temperature bands

Output Hot for 25+, Mild for 15–24, otherwise Cold .

Method / working:

Guided 3 — TRY: Age range using AND

Output Teen ticket only when the age is from 12 to 17 inclusive.

Method / working:

Guided 4 — TRY: Name it and show it

A past OCR programming-techniques question expects the three constructs to be named and supported with a working example . Today: give a C# example of sequence and a C# example of branching . Iteration comes next lesson.

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Positive, negative or zero
Ask the user for an integer. Output exactly one of:
Positive · Negative · Zero

Question 2

Ticket category
Ask for an age and output:
Child for 0–11 · Teen for 12–17 · Adult for 18+

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Delivery charge
Ask for an order total and whether the customer is a member.
Output Free delivery only when the total is at least 30 AND the member answer is "yes" . Otherwise output Delivery charge applies .

Question 4

Simple login
Ask for a username and a PIN.
Output Access granted only when the username is "admin" and the PIN is 2468 .
Otherwise output Access denied .

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

In C#, == is used to compare two values for equality.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Build the missing decision
CODE:

```
Console.Write("Number: ");
int number = int.Parse(Console.ReadLine());

if (number % 2 == 0)
{
Console.WriteLine("Even");
}
else
{
Console.WriteLine("Odd");
}
```

C# % is the remainder operator; it matches OCR MOD .

Guided 2 — CHECK

Which condition comes first?
Test the highest threshold first so 25+ is not caught by the 15+ branch.

Guided 3 — CHECK

Write one Boolean condition

```
if (age >= 12 && age <= 17)
```

Guided 4 — CHECK

Name it and show it
Use the technical vocabulary: sequence, branching/selection, iteration.